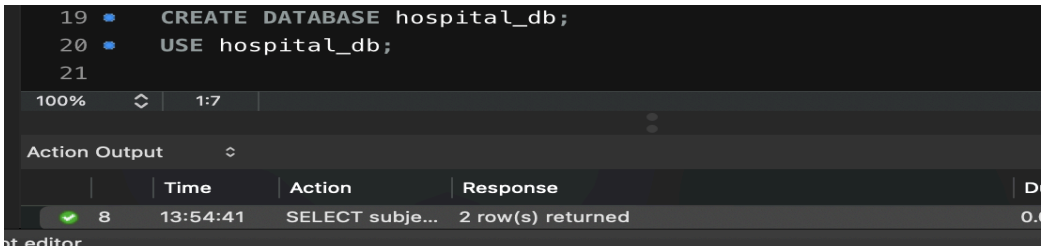
SQL Practice — Physician & Department Database

JOINS, GROUP BY, SUBQUERIES & AGGREGATE FUNCTIONS

Name: K.VAISHALI
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Section: 7

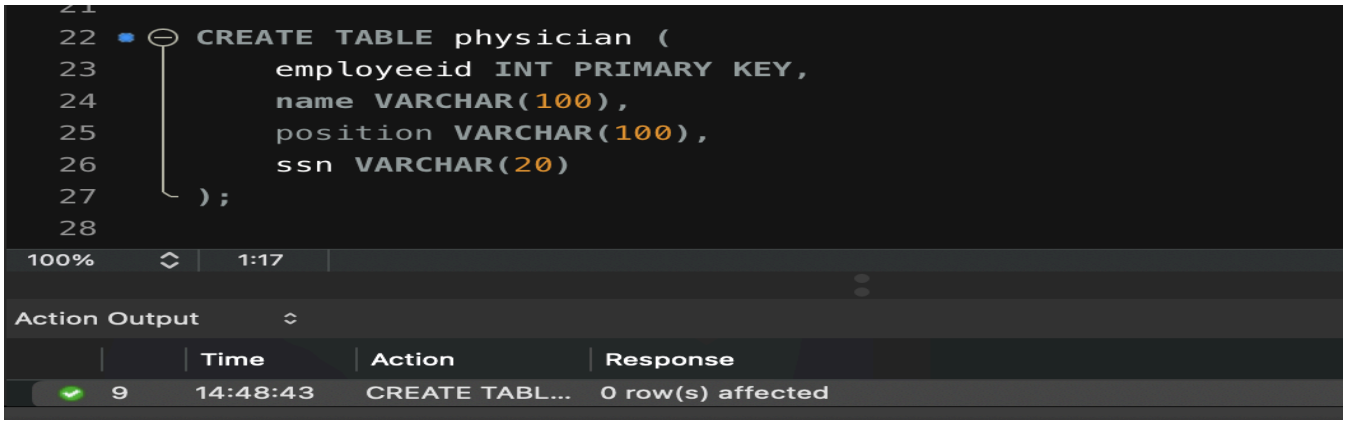
Step 1 — Create Database

```
CREATE DATABASE hospital_db;
USE hospital_db;
```



Step 2 — Create Physician Table (DDL)

```
CREATE TABLE physician (
    employeeid INT PRIMARY KEY,
    name VARCHAR(100),
    position VARCHAR(100),
    ssn VARCHAR(20)
);
```



Step 3 — Insert Data into Physician and Display

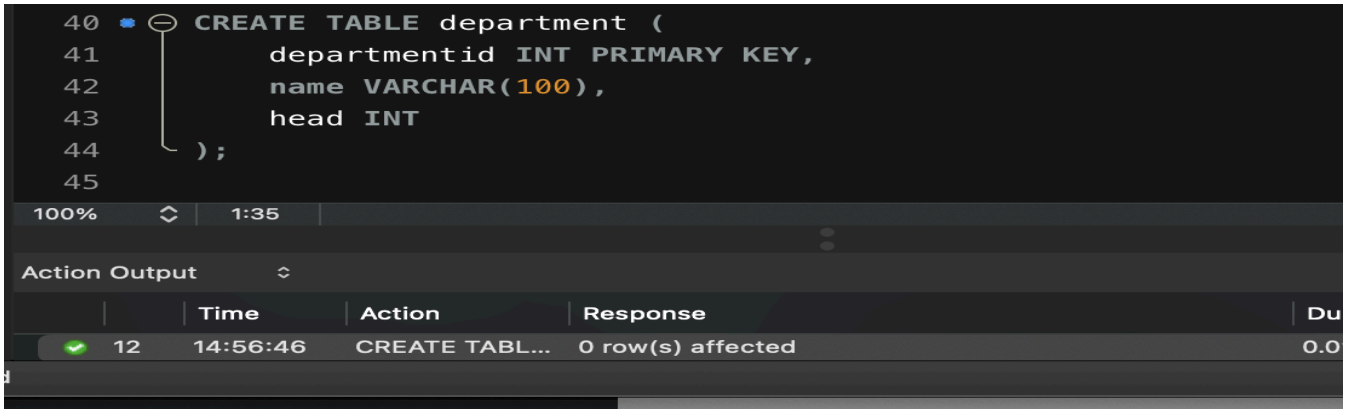
```
INSERT INTO physician VALUES
(1,'Hansika Reddy','Staff Internist','111111111'),
(2,'Priya Sharma','Attending Physician','222222222'),
(3,'Arjun Iyer','Surgical Attending Physician','333333333'),
(4,'Aditya Rao','Senior Attending Physician','444444444'),
(5,'Rohan Mehta','Head Chief of Medicine','555555555'),
(6,'Vikram Nair','Surgical Attending Physician','666666666'),
(7,'Aditya Verma','Surgical Attending Physician','777777777'),
(8,'Karthik Iyer','MD Resident','888888888'),
(9,'Sneha Pillai','Attending Psychiatrist','999999999');

SELECT * FROM physician;
```

employeeid	name	position	ssn
1	Hansika Reddy	Staff Internist	111111111
2	Priya Sharma	Attending Physician	222222222
3	Arjun Iyer	Surgical Attending Physician	333333333
4	Aditya Rao	Senior Attending Physician	444444444
5	Rohan Mehta	Head Chief of Medicine	555555555
6	Vikram Nair	Surgical Attending Physician	666666666
7	Aditya Verma	Surgical Attending Physician	777777777
8	Karthik Iyer	MD Resident	888888888
9	Sneha Pillai	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Step 4 — Create Department Table (DDL)

```
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);
```



Step 5 — Insert Data into Department and Display

```
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);

SELECT * FROM department;
```

1	General Medicine	4	
2	Surgery	7	
3	Psychiatry	9	
NULL	NULL	NULL	

Step 6 — Question 1: INNER JOIN

```
SELECT d.name AS Department,
       p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;
```

	Department	Head_Physician	
	General Medicine	Aditya Rao	
	Surgery	Aditya Verma	
	Psychiatry	Sneha Pillai	

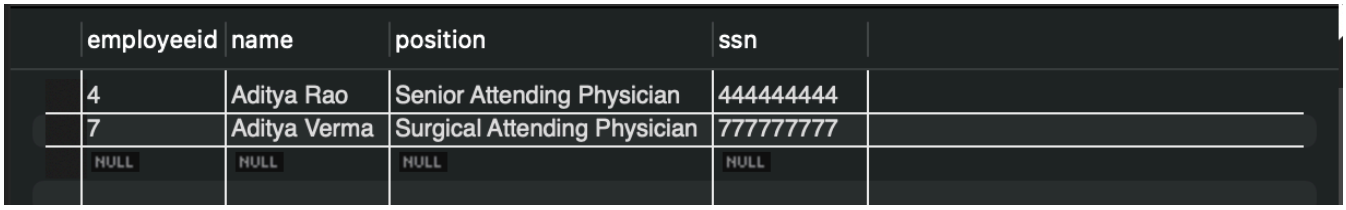
Step 7 — Question 2: WHERE Clause

```
SELECT *
FROM physician
WHERE position='Surgical Attending Physician';
```

	employeeid	name	position	ssn	
	3	Arjun Iyer	Surgical Attending Physician	333333333	
	6	Vikram Nair	Surgical Attending Physician	666666666	
	7	Aditya Verma	Surgical Attending Physician	777777777	
	NULL	NULL	NULL	NULL	

Step 8 — Question 3: WHERE + LIKE

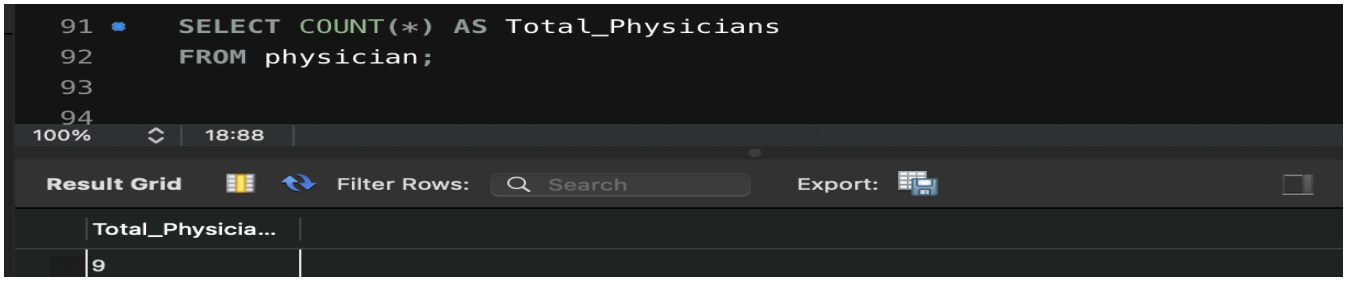
```
SELECT *
FROM physician
WHERE name LIKE 'Aditya%';
```



employeeid	name	position	ssn
4	Aditya Rao	Senior Attending Physician	444444444
7	Aditya Verma	Surgical Attending Physician	777777777
NULL	NULL	NULL	NULL

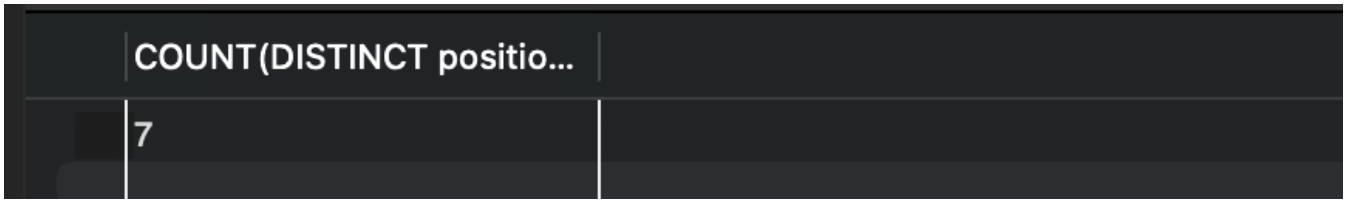
Step 9 — Question 4: COUNT()

```
SELECT COUNT(*) AS Total_Physicians
FROM physician;
```



Step 10 — Question 5: COUNT DISTINCT

```
SELECT COUNT(DISTINCT position)
FROM physician;
```



COUNT(DISTINCT positio...
7

Step 11 — Question 6: GROUP BY

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position;
```

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Step 12 — Question 7: GROUP BY + HAVING

```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

position	Total_Employees
Surgical Attending Physician	3

Step 13 — Question 8: ORDER BY

```
SELECT *
FROM physician
ORDER BY name ASC;
```

employeeid	name	position	ssn
4	Adity…	Senior Attending Physician	444444444
7	Adity…	Surgical Attending Physician	777777777
3	Arjun…	Surgical Attending Physician	333333333
1	Hans…	Staff Internist	111111111
8	Karth…	MD Resident	888888888
2	Priya…	Attending Physician	222222222
5	Roha…	Head Chief of Medicine	555555555
9	Sneh…	Attending Psychiatrist	999999999
6	Vikra…	Surgical Attending Physician	666666666
NULL	NULL	NULL	NULL

Step 14 — Question 9: ORDER BY DESC

```
SELECT *
FROM physician
ORDER BY employeeid DESC;
```

employeeid	name	position	ssn
9	Sneh…	Attending Psychiatrist	999999999
8	Karth…	MD Resident	888888888
7	Adity…	Surgical Attending Physician	777777777
6	Vikra…	Surgical Attending Physician	666666666
5	Roha…	Head Chief of Medicine	555555555
4	Adity…	Senior Attending Physician	444444444
3	Arjun…	Surgical Attending Physician	333333333
2	Priya…	Attending Physician	222222222
1	Hans…	Staff Internist	111111111
NULL	NULL	NULL	NULL

Step 15 — Question 10: WHERE + IN.

```
SELECT *
FROM physician
WHERE employeeid IN
(
    SELECT head
    FROM department
);
```

employeeid	name	position	ssn
4	Adity…	Senior Attending Physician	444444444
7	Adity…	Surgical Attending Physician	777777777
9	Sneh…	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Step 16 — Question 11: Subquery (NOT IN)

```
SELECT *
FROM physician
WHERE employeeid NOT IN
(
    SELECT head
    FROM department
);
```

employeeid	name	position	ssn
1	Hans…	Staff Internist	111111111
2	Priya…	Attending Physician	222222222
3	Arjun…	Surgical Attending Physician	333333333
5	Roha…	Head Chief of Medicine	555555555
6	Vikra…	Surgical Attending Physician	666666666
8	Karth…	MD Resident	888888888
NULL	NULL	NULL	NULL

Step 17 — Question 12: GROUP BY + ORDER BY

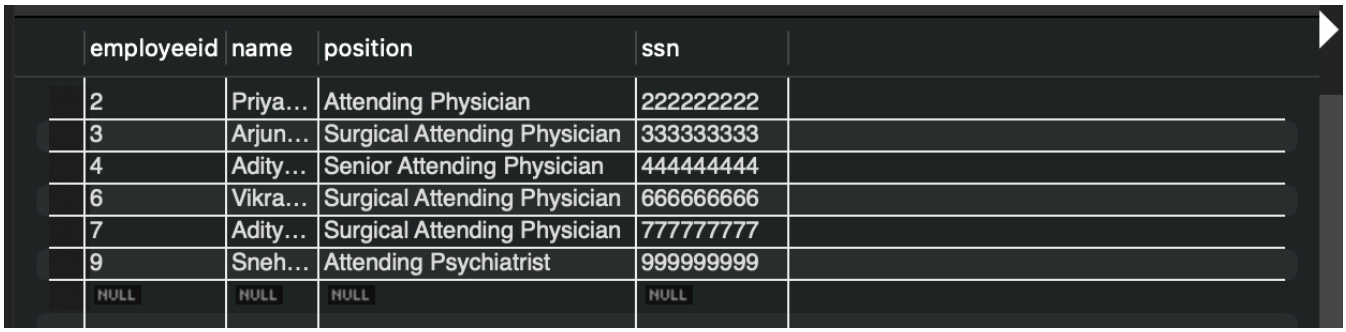
```
SELECT position,
       COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
ORDER BY Total_Employees DESC;
```



position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Step 18 — Question 13: WHERE + LIKE

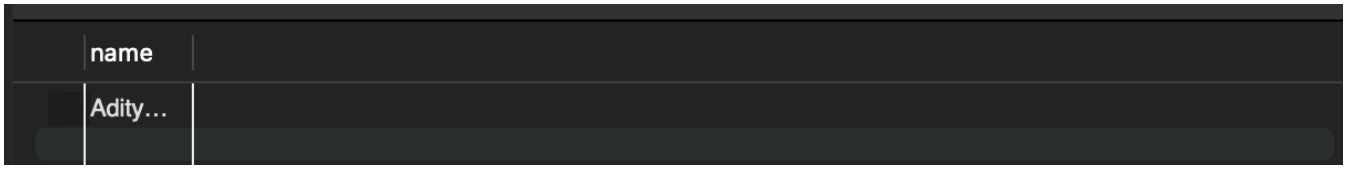
```
SELECT *
FROM physician
WHERE position LIKE '%Attending%';
```



employeeid	name	position	ssn
2	Priya…	Attending Physician	222222222
3	Arjun…	Surgical Attending Physician	333333333
4	Adity…	Senior Attending Physician	444444444
6	Vikra…	Surgical Attending Physician	666666666
7	Adity…	Surgical Attending Physician	777777777
9	Sneh…	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Step 19 — Question 14: JOIN + WHERE

```
SELECT p.name
FROM physician p
JOIN department d
ON p.employeeid=d.head
WHERE d.name='Surgery';
```



name
Adity…

Step 20 — Question 15: JOIN + ORDER BY

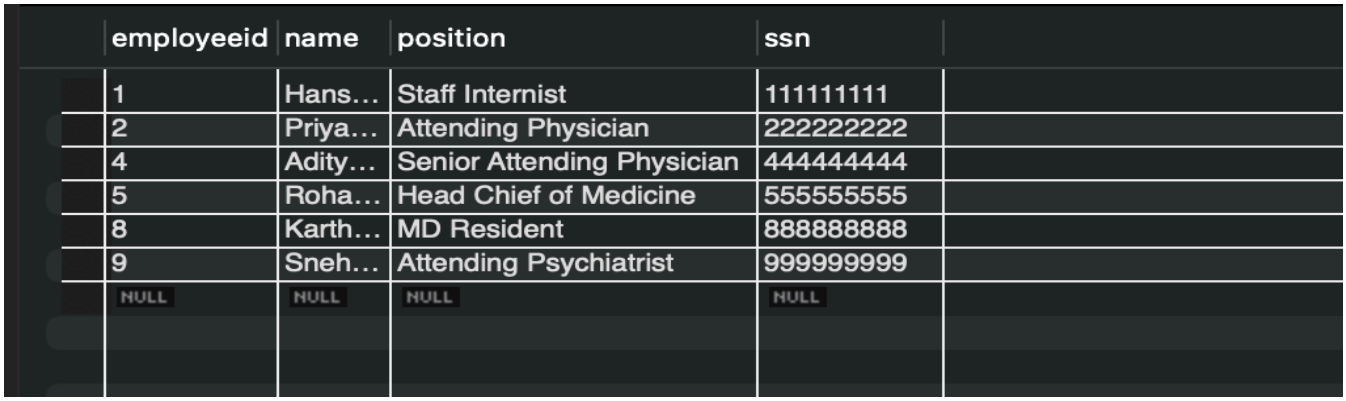
```
SELECT d.name,
       p.name
FROM department d
JOIN physician p
ON d.head=p.employeeid
ORDER BY d.name;
```



name	name
General Medicine	Adity...
Psychiatry	Sneh...
Surgery	Adity...

Step 21 — Question 16: WHERE + NOT LIKE

```
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
```



employeeid	name	position	ssn
1	Hans...	Staff Internist	111111111
2	Priya...	Attending Physician	222222222
4	Adity...	Senior Attending Physician	444444444
5	Roha...	Head Chief of Medicine	555555555
8	Karth...	MD Resident	888888888
9	Sneh...	Attending Psychiatrist	999999999
NULL	NULL	NULL	NULL

Step 22 — Question 17: COUNT + HAVING

```
SELECT position,
       COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
```

Result Grid | Filter Rows: Search | Export:

position	Total
Surgical Attending Physician	3

Step 23 — Question 18: EXISTS.

```
SELECT *
FROM department d
WHERE EXISTS
(
    SELECT 1
    FROM physician p
    WHERE p.employeeid=d.head
);
```

department...	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
NULL	NULL	NULL

Step 24 — Question 19: MAX()

```
SELECT MAX(employeeid)
FROM physician;
```

MAX(employeei...
9

Step 25 — Question 20: MIN()

```
SELECT MIN(employeeid)
FROM physician;
```

MIN(employeei...
1